Signed word composition: source derivation

Signed word bridge: source-derived invariant

State: signs ±1, canonical offsets/anchors, prime 2 < r ≤ 2^63
ID initializes it; recursive find and both union orientations preserve it

Sign table closes all i8 arithmetic; existing Ring proofs close scalar operations
cu+cv ∈ {-2,0,2}; s=-cu·cv ∈ {-1,1}; no signed arithmetic overflow

Odd r = 2h+1: inverse(+2)=h+1; inverse(-2)=h
Reuse checked modular exponentiation + canonical inverse uniqueness

Invalid sign0: edge(x,0,x,1,1,0), r=101, true x=1 → stored anchor50

Fixture modulus bound is entry-point-specific; counters/index validity remain separate

Written derivation and source audit only; no new Lean check, execution or timing

The machine bridge reduces to a finite sign invariant plus existing scalar arithmetic. For odd $r = 2h + 1$, the canonical inverses of $+2$ and -2 are $h+1$ and h. This is a written source-bound derivation, not a new Lean acceptance claim.

Preservation. ID initializes sign1 and zero offsets. Find multiplies signs and composes canonical offsets through signed/add. Union uses $s = -c_u c_v$ and reverses offsets using $-s$; all these signs remain ± 1. The sum $c_u + c_v$ is exactly $-2, 0, 2$, so i8 arithmetic cannot overflow. Existing GraphRingWord lemmas handle canonical add/sub/neg/widened multiplication at $r \leq 2^{63}$. Anchor transport preserves canonical pairs.

Inverse simplification. Since $2(h+1) = r+1$ and $(-2)h = 1-r$, the canonical inverse representatives are $r/2+1$ and $r/2$. Both fit a word without forming $r+1$. Existing generated-inverse correctness and uniqueness identify native modpow output with these values. This makes no change to runtime or counters.

Counterexample to omitted sign premise. Let $r = 101$, true root value $x = 1$, and call edge$(x, 0, x, 1, 1, 0)$. The incoming equation is true, but the source takes the nonzero same-root branch, chooses denominator99, inverse50, and stores anchor$(50, 0)$. Collector sites construct only ± 1; the public edge interface does not enforce that premise.

Caller scope and concrete obligations

Actual source. three_leads.rs1543–1579 initializes signs;1582–1597 composes find;1658–1711 reduces edges and updates union/anchors. Its collector constructs signs at1908–1921. The frozen Kummer Adapter graph.rs11 also constructs only ± 1. The fixture original_gstep_fixture in pruned.rs814 asserts $2 \le r \le 2^{63}$, then asserts canonical labels before doubling. That guard belongs to this fixture, not every entry point in the file.

Bounds. For canonical inputs, $a + b \le 2r - 2 < 2^{64}$ at $r \le 2^{63}$. Above that bound the maximal canonical pair overflows. Noncanonical inputs defeat literal Ring.neg because it subtracts before reducing; signed$(+1, a)$ returns a unchanged. The field theorem therefore needs exact sign and canonical-word premises. Counter/size headroom and valid array indices remain separate.

Next checked interface. Extend mutable graph state with GoodWords and prove native word-Run preservation/refinement using existing scalar lemmas and the four-entry sign table. Extend edge to preserve that predicate in both union orientations and the same-root branch. Then compose decoded state with same_after_two_finds and canonical equality reflection. Specialize existing inversion to the closed half-modulus values. The collector still supplies its geometric true-edge certificate.

Evidence and status. REPORT.md contains the full written induction and precise source mapping. obligations.json records eight interfaces and their present status; source-bindings.sha256 pins inspected files. The GraphRing packet has historical Python-gate provenance and is read as an existing proof dependency; this audit does not re-certify that workflow. No code compilation, formal replay, timing, scientific execution, repository edit or cloud operation occurred.